5.3.7 Improving accuracy via iterative refinement

When solving $Ax = b$ on a computer, error is inherently incurred. Instead of the exact solution x, an approximate solution $\widehat{x}$ is computed, which instead solves $A\widehat{x} = \hat{b}$. The difference between x and $\widehat{x}$ satisfies

$$A(x - \widehat{x}) = b - \hat{b}.$$

We can compute $\hat{b} = A\widehat{x}$ and hence we can compute $\delta b = b - \hat{b}$. We can then solve $A\delta x = \delta b$. If this computation is completed without error, then $x = \widehat{x} + \delta x$ and we are left with the exact solution. Obviously, there is error in δx as well, and hence we have merely computed an improved approximate solution to $Ax = b$. This process can be repeated. As long as solving with A yields at least one digit of accuracy, this process can be used to improve the computed result, limited by the accuracy in the right-hand side b and the condition number of A.

This process is known as iterative refinement.